

Your grade: **100%**

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. What is 4! ?

1 / 1 point

- ☐ 10
- ☒ 24
- ☐ 8
- ☐ 6

✔ Nice work

This is correct. $4!=1*2*3*4=24$

2. What is P(5,2)?

1 / 1 point

- ☐ 120
- ☐ 60
- ☒ 20
- ☐ 10

✔ Nice work

This is correct. $P(5,2)=5!/(5-2)!=20$

3. How many non-repetitive words of length 4 can you make from the letters {a, b, c, d, e, f}?

1 / 1 point

Note: aabc is not accepted because 'a' occurs more than once.

- ☐ 30
- ☒ 360
- ☐ 720
- ☐ 60

✔ Nice work

This is correct. It is $P(6,4)=6!/(6-4)!=360$.

4. How many non-repetitive words of length 6 can you make from the letters {a, o, c, d, e, f, g, h, i} containing 'dog'?

1 / 1 point

Note: odogie is not accepted because 'o' occurs more than once and edcogh is not accepted because 'dog' does not occur as a block.

- ☒ 480
- ☐ 120
- ☐ 60480
- ☐ 840

✔ Nice work

This is correct. Since ‘dog’ has three letters and we are looking for words containing ‘dog’ with length 6, it means that we can only select three more letters from the six remaining letters, which is $P(6,3)=6!/(6-3)!=120$. But considering ‘dog’ as one block, it can be seen at position 1, 2, 3 or 4. So, we have four cases, each of which can generate 120 words, which is 480 words in total.